Lagrange Duality (0.5 pt)

Consider the optimization problem

$$\begin{array}{ll} \text{minimize} & \underbrace{x^2+1}_{f_0} \\ \text{subject to} & \underbrace{(x-2)(x-4)}_{f_1} \le 0 \end{array}$$

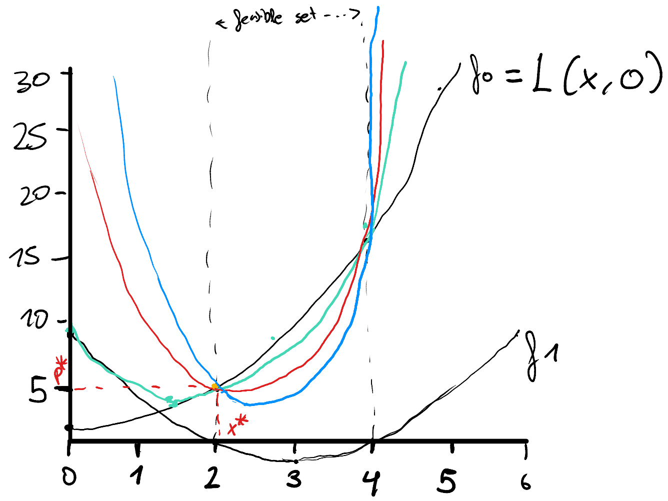
with variable $x \in \mathbb{R}$.

(a) Analysis of primal problem: give the feasible set, the optimal value, and the optimal solution.

The constraint $(x-2)(x-4)\le 0$ is true in the interval $[2,4]$.
Then $\Omega = [2,4]$.
As the objective x^2+1 is increasing, the minimum occurs at the minimum feasible value for x. The optimal solution is therefore $f_0(2)=2^2+1=5$, using the optimal value 2. $x^*=2$, $p^*=5$

(b) Lagrangian and dual function: plot the objective x^2+1 versus x. On the same plot, show the feasible set, optimal point and value, and plot the Lagrangian $L(x,\lambda)$versus x for a few positive values of λ. Verify the lower bound property ($p^* \ge inf_x L(x,\lambda)$ for $\lambda \ge 0$). Compute and sketch the Lagrange dual function g.

$$L(x,\lambda) = x^2+1+\lambda(x-2)(x-4) = x^2+1+\lambda(x^2-6x+8) =$$
$$= (1+\lambda)x^2 - 6\lambda x + 1 + 8\lambda$$



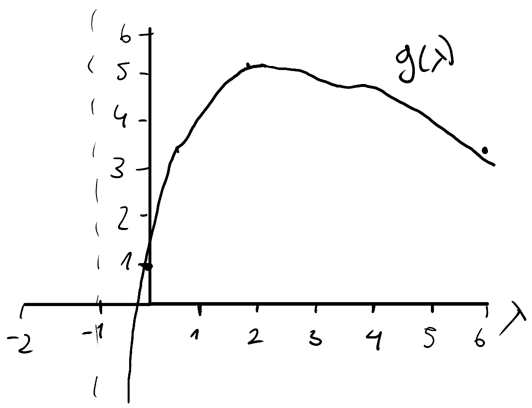
$L(x,1) = 2x^2 - 6x + 9$
$L(x,2) = 3x^2 - 12x + 17$
$L(x,3) = 4x^2 - 18x + 25$

We see that the lower bound property holds. For $\lambda = 2$, $p^* = \inf_x L(x,\lambda)$.
Otherwise $p^* \ge \inf_x L(x,\lambda)$.

$$L(x,\lambda) = (1+\lambda)x^2 - 6\lambda x + (1+8\lambda).$$

$$g(\lambda) = \inf_x L(x,\lambda) = L\left(\tfrac{3\lambda}{1+\lambda},\lambda\right) = \cancel{(1+\lambda)}\frac{(3\lambda)^2}{(1+\lambda)^{\cancel{2}}} - 6\lambda\frac{3\lambda}{1+\lambda} + 1 + 8\lambda = \frac{-9\lambda^2}{1+\lambda} + 8\lambda + 1$$

↳ $L'(x,\lambda) = 0 \Rightarrow 2(1+\lambda)x - 6\lambda = 0 \Rightarrow x = \frac{3\lambda}{1+\lambda}$



So $g(\lambda) = \begin{cases} \frac{-9\lambda^2}{1+\lambda} + 8\lambda + 1 & \lambda > -1 \\ -\infty & \lambda \le -1 \end{cases}$

(c) Lagrange dual problem: state the dual problem, and verify that it is a concave maximization problem. Find the dual optimal value and dual optimal solution λ. Does strong duality hold?

The Lagrange dual problem is:

$$\begin{array}{ll} \text{maximize} & \frac{-9\lambda^2}{1+\lambda} + 8\lambda + 1 \\ \text{subject to} & \lambda \geq \end{array}$$

It is concave as it is visible in the graph above. The optimal solution is $\lambda = 2$, so $d^* = g(2) = 5$.
As $p^* = d^*$ strong duality holds.

KKT Condition (0.5 pt)

Consider the following optimization problem:

$$\begin{array}{ll} \text{minimize} & x_1^2 - 2x_2^2 \\ \text{subject to} & (x_1+4)^2 - 2 \leq x_2 \\ & x_1 - x_2 + 4 = 0 \\ & x_1 \geq -10 \end{array}$$

(1) sketch the problem and graphically determine the primal solution x^*.

We can transform the problem by eliminating the liner equality $x_1 - x_2 + 4 = 0$.
$x_2 = x_1 + 4$

objective:

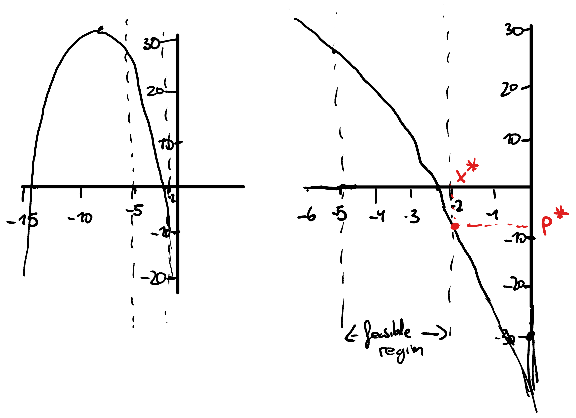
$$x_1^2 - 2x_2^2 = x_1^2 - 2(x_1+4)^2 = x_1^2 - 2(x_1^2 + 8x_1 + 16) =$$
$$= -x_1^2 - 16x_1 - 32$$

inequality:

$$(x_1+4)^2 - 2 \leq x_2$$
$$(x_1+4)^2 - 2 - x_1 - 4 \leq 0 \Rightarrow x_1^2 + 8x_1 + 16 - 2 - x_1 - 4 = x_1^2 + 7x_1 + 10 = (x+5)(x+2) \leq 0$$

So we have simplified the problem to:

$$\begin{array}{ll} \text{minimize} & -x^2 - 16x - 32 \\ \text{subject to} & (x+5)(x+2) \leq 0 \\ & -x - 10 \leq 0 \end{array}$$



feasible set betwen the roots, $x \in [-5, -2]$.
Objective is decreasing in that interval so x^* must be at the end of the interval $x^* = -2$. Then $p^* =$

$$= -4 + 32 - 32 = -4.$$

So the primal solution is:
$x^* = (-2, 2)$

(2) verify your x^* by determining suitable λ^* and ν^* such that the KKT conditions are satisfied for (x^*, λ^*, ν^*).

$$L(x_1, x_2, \lambda_1, \lambda_2, v) = x_1^2 - 2x_2^2 + \lambda_1\left((x_1+4)^2 - 2 - x_2\right) + \lambda_2(-x_1 - 10) + v(x_1 - x_2 + 4)$$

1.- Primal constraints: $(x_1+4)^2 - 2 - x_2 \leq 0$

$-x_1 - 10 \leq 0$

$x_1 - x_2 + 4 = 0$

2.- Dual constraints: $\lambda_1, \lambda_2 \geq 0$

3.- Complementary slackness: $\lambda_1\left((x_1+4)^2 - 2 - x_2\right) = \lambda_2(-x_1 - 10) = 0$

4.- Vanishing gradient of Lagrangian

$$\frac{\partial L}{\partial x_1} = 2x_1 + 2\lambda_1(x_1+4) - \lambda_2 + v = 0$$

$$\frac{\partial L}{\partial x_2} = -4x_2 - \lambda_1 - v = 0$$

$\boxed{\lambda_2 = 0}$ because $-x_1 - 10 \leq 0$ is inactive at $(-2, 2)$.

Gradient at $(-2, 2)$:

w.r.t x_1: $-4 + 4\lambda_1 + v = 0 \Rightarrow v = 4 - 4\lambda_1$

w.r.t x_2: $-8 - \lambda_1 - v = -8 - \lambda_1 - 4 + 4\lambda_1 = -12 + 3\lambda_1 = 0 \Rightarrow \boxed{\lambda_1 = 4}$

$\boxed{v = 4 - 4 \cdot 4 = -12}$

All KKT conditions are satisfied for:

$x^* = (-2, 2)$, $\lambda_1^* = 4$, $\lambda_2^* = 0$, $v^* = -12$.